LOCALIZED CHERN CLASSES OF A DERIVED TENSOR PRODUCT ARE GIVEN BY THE UNIVERSAL POLYNOMIAL FORMULA IN THE TWO LOCALIZED CLASSES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FBF (sources/stacks-chow/pinned-original-source.tex line 11525).

Allowed standard prerequisites (editorial).

- Identification of the localized classes with those computed from a blowing up. Same chapter, Lemma labelled lemma-independent-loc-chern-bQ, cited at proof lines 11586-11592 and 11604-11618. Allows both F_i to be handled with one common W, b and pair of complexes Q_i.
- The polynomial expansion of c'_p and P'_p for a tensor product over a common closed subscheme. Same chapter, Lemma labelled lemma-silly-tensor-product, cited at proof lines 11634-11636. Expands the localized classes of Q_1 otimes^L Q_2 restricted to E_1 cap E_2.
- Agreement of polynomials in restricted and unrestricted localized classes. Same chapter, Lemma labelled lemma-homomorphism-final, cited at proof lines 11636-11638. Converts the identity on E_1 cap E_2 into the identity on Z_1 cap Z_2.
- Assumption (3) of the localized Chern situation for perfect objects. Same chapter, Situation labelled situation-loc-chern, invoked in the statement. Guarantees that all the localized classes occurring in the formula are defined.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Same-chapter pointer into the author environment situation at source line 1372; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

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Source reference. Original reference 0GUJ: sources/stacks-chow/pinned-original-source.tex, source lines 10678–10678. Same-chapter pointer into the author environment situation at source line 10678; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FE8: sources/stacks-chow/pinned-original-source.tex, source lines 11165–11165. Same-chapter pointer into the author environment lemma at source line 11165; invoked at proof line 11571 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FE5: sources/stacks-chow/pinned-original-source.tex, source lines 9958–9958. Same-chapter pointer into the author environment lemma at source line 9958; invoked at proof line 11630 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FE7: sources/stacks-chow/pinned-original-source.tex, source lines 11026–11026. Same-chapter pointer into the author environment lemma at source line 11026; invoked at proof line 11634 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In Situation 02QL let X be locally of finite type over S. Let $Z_i \subset X$, $i = 1, 2$ be closed subschemes. Let F_i, $i = 1, 2$ be perfect objects of $D(\mathcal{O}_X)$. Assume for $i = 1, 2$ that $F_i|_{X \setminus Z_i}$ is zero*[1] *and that F_i on X satisfies assumption (3) of Situation 0GUJ. Denote $r_i = P_0(Z_i \to X, F_i) \in A^0(Z_i \to X)$. Then we have*

$$c_1(Z_1 \cap Z_2 \to X, F_1 \otimes^{\mathbf{L}}_{\mathcal{O}_X} F_2) = r_1 c_1(Z_2 \to X, F_2) + r_2 c_1(Z_1 \to X, F_1)$$

in $A^1(Z_1 \cap Z_2 \to X)$ and

$$\begin{aligned} c_2(Z_1 \cap Z_2 \to X, F_1 \otimes^{\mathbf{L}}_{\mathcal{O}_X} F_2) = {} & r_1 c_2(Z_2 \to X, F_2) + r_2 c_2(Z_1 \to X, F_1) + \\ & \binom{r_1}{2} c_1(Z_2 \to X, F_2)^2 + \\ & (r_1 r_2 - 1) c_1(Z_2 \to X, F_2) c_1(Z_1 \to X, F_1) + \\ & \binom{r_2}{2} c_1(Z_1 \to X, F_1)^2 \end{aligned}$$

in $A^2(Z_1 \cap Z_2 \to X)$ and so on for higher Chern classes. Similarly, we have

$$ch(Z_1 \cap Z_2 \to X, F_1 \otimes^{\mathbf{L}}_{\mathcal{O}_X} F_2) = ch(Z_1 \to X, F_1) ch(Z_2 \to X, F_2)$$

in $\prod_{p \geq 0} A^p(Z_1 \cap Z_2 \to X) \otimes \mathbf{Q}$. More precisely, we have

$$P_p(Z_1 \cap Z_2 \to X, F_1 \otimes^{\mathbf{L}}_{\mathcal{O}_X} F_2) = \sum\nolimits_{p_1 + p_2 = p} \binom{p}{p_1} P_{p_1}(Z_1 \to X, F_1) P_{p_2}(Z_2 \to X, F_2)$$

in $A^p(Z_1 \cap Z_2 \to X)$.

Proof. Choose proper morphisms $b_i : W_i \to \mathbf{P}^1_X$ and $Q_i \in D(\mathcal{O}_{W_i})$ as well as closed subschemes $T_i \subset W_{i,\infty}$ as in the construction of the localized Chern classes for F_i or more generally as in Lemma 0FE8. Choose a commutative diagram

$$\begin{array}{ccc} W & \xrightarrow{g_2} & W_2 \\ \downarrow{\scriptstyle g_1} & \overset{b}{\searrow} & \downarrow{\scriptstyle b_2} \\ W_1 & \xrightarrow{b_1} & \mathbf{P}^1_X \end{array}$$

[1] Presumably there is a variant of this lemma where we only assume $F_i|_{X \setminus Z_i}$ is isomorphic to a finite locally free $\mathcal{O}_{X \setminus Z_i}$-module of rank $< p_i$.

where all morphisms are proper and isomorphisms over $\mathbf{A}^1_X$. For example, we can take W to be the closure of the graph of the isomorphism between $b_1^{-1}(\mathbf{A}^1_X)$ and $b_2^{-1}(\mathbf{A}^1_X)$. By Lemma 0FE8 we may work with W, $b = b_i \circ g_i$, $Lg_i^*Q_i$, and $g_i^{-1}(T_i)$ to construct the localized Chern classes $c_p(Z_i \to X, F_i)$. Thus we reduce to the situation described in the next paragraph.

Assume we have

(1) a proper morphism $b : W \to \mathbf{P}^1_X$ which is an isomorphism over $\mathbf{A}^1_X$,
(2) $E_i \subset W_\infty$ is the inverse image of Z_i,
(3) perfect objects $Q_i \in D(\mathcal{O}_W)$ whose Chern classes are defined, such that
   (a) the restriction of Q_i to $b^{-1}(\mathbf{A}^1_X)$ is the pullback of F_i, and
   (b) there exists a closed subscheme $T_i \subset W_\infty$ containing all points of W_∞ lying over $X \setminus Z_i$ such that $Q_i|_{T_i}$ is zero.

By Lemma 0FE8 we have

$$c_p(Z_i \to X, F_i) = c'_p(Q_i) = (E_i \to Z_i)_* \circ c'_p(Q_i|_{E_i}) \circ C$$

and

$$P_p(Z_i \to X, F_i) = P'_p(Q_i) = (E_i \to Z_i)_* \circ P'_p(Q_i|_{E_i}) \circ C$$

for $i = 1, 2$. Next, we observe that $Q = Q_1 \otimes^{\mathbf{L}}_{\mathcal{O}_W} Q_2$ satisfies (3)(a) and (3)(b) for $F_1 \otimes^{\mathbf{L}}_{\mathcal{O}_X} F_2$ and $T_1 \cup T_2$. Hence we see that

$$c_p(Z_1 \cap Z_2 \to X, F_1 \otimes^{\mathbf{L}}_{\mathcal{O}_X} F_2) = (E_1 \cap E_2 \to Z_1 \cap Z_2)_* \circ c'_p(Q|_{E_1 \cap E_2}) \circ C$$

and

$$P_p(Z_1 \cap Z_2 \to X, F_1 \otimes^{\mathbf{L}}_{\mathcal{O}_X} F_2) = (E_1 \cap E_2 \to Z_1 \cap Z_2)_* \circ P'_p(Q|_{E_1 \cap E_2}) \circ C$$

by the same lemma. By Lemma 0FE5 the classes $c'_p(Q|_{E_1 \cap E_2})$ and $P'_p(Q|_{E_1 \cap E_2})$ can be expanded in the correct manner in terms of the classes $c'_p(Q_i|_{E_i})$ and $P'_p(Q_i|_{E_i})$. Then finally Lemma 0FE7 tells us that polynomials in $c'_p(Q_i|_{E_i})$ and $P'_p(Q_i|_{E_i})$ agree with the corresponding polynomials in $c'_p(Q_i)$ and $P'_p(Q_i)$ as desired. □